Leveling Up Aptitude: A Gamified Educational Intervention for Entrance Exams

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***Abstract*—While standardized entry tests play a crucial role in selecting the academic and career paths of individuals, they can pose substantial obstacles to students from varied backgrounds. This study explores the possibility of using gamification in education to improve exam-taking skills and address these issues. The study employs a structured program that is based on gamification concepts and aims to include participants in interactive learning experiences. The study assesses the effect of the intervention on participants' aptitude levels, cognitive abilities, and motivation using a combination of theoretical frameworks and empirical data. Using a mixed-methods approach, the study combines qualitative insights with quantitative assessments. Pre- and post-intervention assessments are used in quantitative analysis to track changes in participants' cognitive abilities and aptitude levels. In the meantime, participant questionnaires and interviews are used to collect qualitative data, which offers more in-depth understanding of their opinions and experiences with the gamified learning strategy. The study provides a thorough understanding of the effectiveness of the intervention and its implications for educational practice by triangulating these methodologies. The results show that participants' aptitude levels and cognitive abilities are positively impacted by the gamified educational intervention. Furthermore, the gamified activities' immersive and interactive elements are credited by participants for their increased motivation and involvement with the learning process. These findings highlight gamification's potential as a tactic to improve academic achievement gaps and improve educational outcomes. In order to support various learners on their academic journeys, the paper advocates for the incorporation of gamified approaches into standardized test preparation. It finishes with thoughts on the broader implications for educational practice and policy.**

Keywords— gamified, educational, qualitative, learning, elements

I. Introduction

Standardized entry tests act as gatekeepers to higher education and professional prospects in today's fiercely competitive educational environment. These tests, which are frequently used to evaluate knowledge, aptitude, and critical thinking abilities, are very important in deciding who gets into elite schools and what career paths they can choose. They do, however, also present substantial difficulties, especially for pupils from disadvantaged families or those with little access to resources. The differences in test-taking strategies and outcomes amplify already-existing inequities in educational attainment and social mobility, underscoring the necessity of creative solutions to level the playing field. Passive learning techniques, practice drills, and rote memorization are common components of traditional exam preparation strategies. Although these approaches might provide small gains, they frequently fall short of involving students or encouraging a deeper comprehension and critical thinking abilities. Furthermore, they might favor students who can afford private tutors or coaching programs disproportionately, which would increase the divide between kids from rich backgrounds and those from underprivileged one. Acknowledging these drawbacks, scholars and educators have been more inclined to use gamification and other alternative educational strategies to improve student learning results and advance fair access to education. Gamification presents a possible paradigm for enhancing the learning process by utilizing game-design aspects and concepts in non-gaming environments. Gamified learning environments incorporate features like leaderboards, points, badges, and immersive narratives to improve student motivation, engagement, and retention. Gamification offers a chance to design dynamic, interactive learning experiences that accommodate a range of learning preferences and styles in the context of preparing for standardized tests. Learners can track their progress, get immediate reward, and iteratively improve their abilities in a fun and encouraging atmosphere with the help of game elements and feedback mechanisms. This study investigates the possibilities of a gamified educational intervention as a cutting-edge strategy for improving entrance exam performance. Based on ideas of motivation, cognitive psychology, and game design, the intervention aims to improve learning results and lessen the drawbacks of conventional test-preparation techniques by utilizing the natural appeal of games. The intervention seeks to develop a sense of autonomy, mastery, and purpose among participants by incorporating components of challenge, progression, and reward. This is intended to produce a pleasant and empowering learning environment. By conducting a methodical assessment of the intervention's influence on participants' cognitive abilities, motivation, and aptitude levels, this study adds to the expanding collection of research on gamified learning and its uses in educational settings. The goal of the research is to remove structural obstacles to academic success in order to advance social justice and educational equity. Through focusing on entrance exams, which act as doors to more advanced study and professional prospects, the intervention hopes to enable students from a variety of circumstances to reach their full potential and follow their dreams of education. Furthermore, the study aims to influence evidence-based practices and policies targeted at enhancing educational outcomes and reducing achievement gaps by examining the efficacy of gamification in this particular setting. The results of this study aim to guide the development and execution of future interventions that prioritize inclusion, accessibility, and efficacy in test preparation and beyond through partnership with educators, policymakers, and community stakeholders.

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II. Literature Review

Millions of students worldwide rely on standardized entry examinations to shape their educational and career paths. Examples of these exams are the SAT and ACT in the United States, the GRE for graduate school admissions, and the GMAT for business school admissions. These tests function as doorkeepers to rich professional prospects and higher education institutions by evaluating aptitude, critical thinking abilities, and subject knowledge [1]. Nonetheless, there has been opposition to the use of standardized testing, arguing that it favors pupils from wealthy families and maintains socioeconomic disparities [2]. Traditional test-preparation strategies frequently include rote memorization, practice exercises, and passive learning strategies. While these strategies may produce temporary improvements, they are not effective in developing critical thinking or deep understanding [3]. Moreover, low-income students face access obstacles due to the high expense of test preparation materials and tutoring services, which exacerbates academic achievement gaps[4]. Consequently, there is an increasing demand for creative treatments that might create fair and equal opportunities for all kids to achieve well on standardized tests. Gamified learning is one such technique that aims to improve motivation, engagement, and learning outcomes by incorporating game-design features and concepts into educational situations [5]. Gamified learning environments leverage the intrinsic attraction of games to create dynamic and interactive

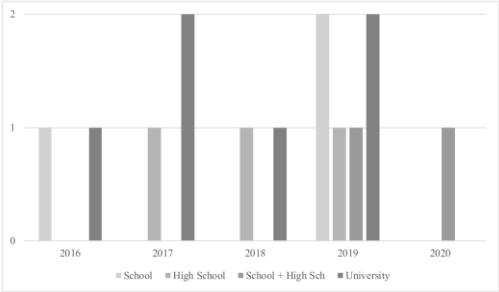

Fig1. number of chosen articles produced annually according to educational level

experiences that pique learners' interest and encourage active interaction [6]. Gamified treatments, which incorporate organized challenges, quick feedback, and possibilities for mastery, have been demonstrated to cultivate a sense of agency and autonomy in students, enabling them to assume responsibility for their own learning process [7]. Gamified learning has been shown to be beneficial in a variety of academic contexts and topic areas by research. A meta-analysis conducted by Hamari et al. (2014), for instance, discovered that gamification has a beneficial impact on learning outcomes, motivation, and engagement. Furthermore, substantial impacts were reported across a range of educational intervention types [8]. In a similar vein, Landers and Landers' (2014) study found that gamified learning environments promote student engagement and motivation, which in turn improves learning outcomes and information retention [9]. Gamified interventions present a potentially effective way to help students prepare for tests by making them more game-like. These treatments allow students opportunity to learn and apply fundamental ideas in a contextually appropriate way by embedding instructional information into immersive and interactive game environments [10]. Moreover, the utilization of game mechanics, such points, badges, and leaderboards, can encourage involvement and recognize advancement, inspiring learners to actively participate in the educational process [11]. Empirical studies on the usefulness of gamified learning in exam preparation are still scarce, despite the potential advantages of this approach. There is a dearth of research that has thoroughly assessed how gamified therapies affect students' performance on standardized tests or their preparedness for admission to graduate or college programs \[12]. Furthermore, the research that has already been done frequently lacks methodological rigor and relies on anecdotal evidence or small sample numbers to bolster assertions regarding the effectiveness of gamified learning for exam preparation [13]. This study attempts to fill these gaps in the literature by conducting a thorough assessment of the usefulness of a gamified educational intervention for improving entrance exam ability. The study uses a mixed-methods approach to investigate how the intervention affects students' aptitude levels, cognitive skills, and motivation. Pre- and post-intervention evaluations, participant surveys, and interviews are all included [14]. The research endeavors to furnish a thorough comprehension of the effectiveness of the intervention and its consequences for educational practice and policy through the process of triangulating data from many sources [15].The results of this study can help shape evidence-based strategies and interventions that support educational equity and help students get ready for standardized tests through cooperation with educators, legislators, and community stakeholders [16]. The research augments current endeavors to establish inclusive and accessible learning environments that enable every student to achieve success by emphasizing the potential of gamification as a pedagogical tool for improving learning outcomes and inspiring students [17].An experimental study was carried out by Sailer, Hense, Mayr, and Mandl (2017) to investigate how particular game design components affect psychological need satisfaction in gamified environments. According to their research, participants' needs for autonomy, competence, and relatedness were strongly impacted by features like points, badges, leaderboards, and progress feedback. This study emphasizes the significance of taking psychological demands into account when designing gamified systems and offers insightful information on the fundamental mechanisms via which gamification drives people [18].A thorough assessment was carried out by Seaborn and Fels (2015) to investigate the theoretical underpinnings and real-world uses of gamification. Their review summarized the body of research on gamification that has been done in a variety of fields, such as business, healthcare, and education, and it focused on important ideas, frameworks, and design principles. This study gives helpful advice for academics and practitioners looking to use gamification in a variety of situations by giving an overview of the state of the field's current gamification research [19]. Landers (2014) established a theoretical framework that connects gamification and serious games to explain gamified learning. In order to create a cohesive theory of gamified learning, Landers combined previous studies using ideas from psychology, education, and game design. This paradigm examines the relationships between the essential elements of gamification—goals, rules, feedback, and challenge—in educational environments. This approach offers a nuanced view of how gamification might improve learning outcomes by envisioning gamified learning as a continuum between serious games and gamification [20].An empirical study was carried out by Hamari and Koivisto

(2015) to look at the social impact of workout games on user behavior. Their study concentrated on the phenomena known as "working out for likes," in which people work out in order to gain acceptance or attention from others. The study clarified the function of social influence in encouraging behavior change by analyzing the effects of social feedback mechanisms in workout games. The design of gamified systems that make use of social components to boost user motivation and engagement will be affected by these findings [21]. Collectively, these studies deepen our knowledge of gamification by examining its theoretical underpinnings, psychological mechanisms, and real-world applications. The initial study emphasizes the significance of taking user motivations into account when designing gamified systems by examining the effects of particular game design components on psychological need satisfaction. The second study synthesizes current literature to identify essential concepts and frameworks, offering a thorough overview of gamification philosophy and practice. In the third study, a theoretical framework for comprehending gamified learning is put forth, providing insights into how gamification and serious games interact in educational settings. The fourth study, which looks at how gamified systems affect user behavior socially, emphasizes the importance of social feedback mechanisms in promoting involvement and engagement.All things considered, these studies add to the expanding corpus of knowledge on gamification by providing insightful information about how it can improve learning outcomes, motivation, and engagement in a variety of settings. Researchers and practitioners may create more meaningful and successful gamified systems that encourage positive behavior change and improve user experiences by knowing the psychological mechanics, theoretical underpinnings, and real-world applications of gamification.

III. Gamified Approaches to Entrance Exam Preparation

In the context of standardized testing, gamified entrance exam preparation methods offer a fresh and creative way to motivate students, engage them, and improve learning outcomes. These methods try to make exam preparation—which may be a daunting task—more interactive and interesting by incorporating game design aspects into educational events. Using game principles to reward and incentivise progress—such as points, badges, leaderboards, and levels—is a fundamental component of gamified techniques. For instance, finishing practice questions, mastering particular topic areas, or hitting particular study plan objectives could reward participants points or badges. These incentives provide participants instant feedback and encouragement, encouraging them to keep up their efforts and monitor their development over time. Furthermore, social engagement and competitive components are frequently incorporated into gamified techniques, enabling participants to compare their performance with peers and engage in friendly rivalry. For example, leaderboards can show who the best performers are in the group, encouraging competition and solidarity amongst players. Participants are more driven and accountable because of the social component, which encourages them to surpass their colleagues and go up the ranks. Gamified entrance exam preparation strategies frequently include immersive narratives and storytelling components in addition to gaming mechanics to contextualize learning exercises and establish a sense of progression. For instance, participants might set out on a virtual journey or adventure, where they advance toward their ultimate objective of passing the admission exam with each practice question or idea they grasp. Gamified techniques improve learning activities' engagement and give the process a feeling of direction and significance by enclosing them in an engaging story. Moreover, gamified methods offer chances for customization and independence, enabling learners to customize their educational journey according to their own requirements and inclinations. It may be up to the individual participants to decide on their own learning route, pick interesting subjects, and establish their own objectives and difficulties. This autonomy encourages intrinsic motivation and gives participants the power to take charge of their learning process, which eventually results in more engagement and consistent effort. All things considered, gamified entrance exam preparation strategies present a viable way to overcome the drawbacks of standardized testing while improving educational results. These methods generate dynamic and immersive learning experiences that capture learners' attention and encourage active engagement by utilizing the motivational power of games and including elements of autonomy, social interaction, competition, and story.

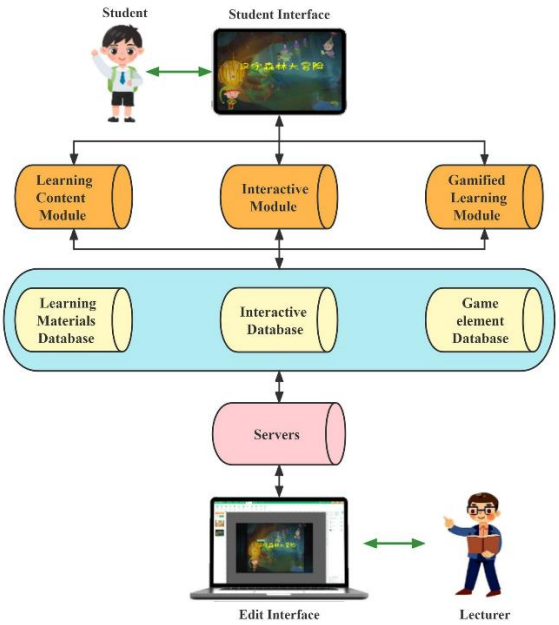

Fig2. Interactive model for learning

Gamified approaches have the potential to completely change how we prepare for admission tests and democratize access to education for all students, as long as research in this area continues to advance. Furthermore, gamified entrance test preparation strategies can successfully cater to participants' varied learning preferences and demands. Not all students will benefit from traditional exam preparation techniques, which frequently focus on rote memory and passive learning. Gamified techniques, on the other hand, provide a range of interactive and multimodal learning experiences that accommodate various learning preferences and styles. For instance, interactive animations and visuals could be helpful for visual learners, whereas audio-based discussions and explanations might be more appealing for auditory learners. Gamified techniques guarantee that participants can interact with the content in ways that are most efficient and relevant for them by offering a variety of learning modalities. Gamified methods also have the benefit of encouraging tenacity and

fortitude in the face of difficulties. The process of preparing for an entrance exam can be difficult and time-consuming, requiring patience and persistent work over a long period of time. Setbacks are reframed as chances for growth and improvement in a friendly and encouraging atmosphere that gamified techniques offer their participants. By providing opportunities for introspection, feedback mechanisms, and gradual challenges, participants can cultivate a growth mindset and accept challenges as a necessary component of learning. Furthermore, by encouraging active learning and in-depth comprehension, gamified approaches can improve knowledge transfer and long-term retention. Through problem-solving, critical thinking, and decision-making exercises, participants actively engage with the material rather than passively receiving it. Participants get a deeper comprehension of important concepts and principles by applying their knowledge in scenarios that are contextually relevant.

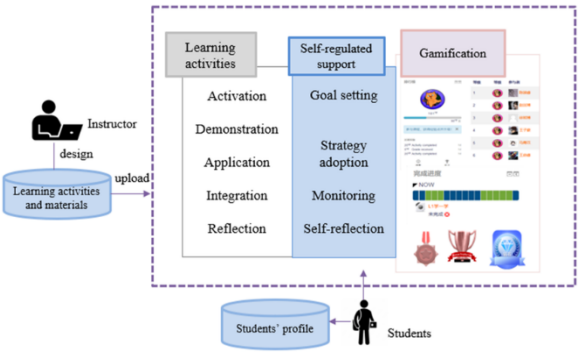

Fig3. The atmosphere and structure of the self-regulated gamified learning system

This understanding is crucial for success on standardized tests and beyond. Furthermore, gamified techniques' immersive and interactive qualities encourage emotional engagement and personal investment in the learning process, which results in more significant and unforgettable learning opportunities. Gamified methods of preparing for admission exams present a fresh and inventive response to the problems associated with standardized testing. These methods produce captivating and immersive learning experiences that meet the requirements and preferences of a wide range of learners by utilizing the motivational power of games and including elements of autonomy, social interaction, story, and competition. Furthermore, gamified methods encourage perseverance, fortitude, and in-depth comprehension, which enhances learning results and prolongs knowledge retention. Gamified techniques hold promise for changing the way we prepare for admission tests and ensuring equitable access to educational opportunities for all learners as educators and researchers continue to examine the potential of gamification in education.

IV. Addressing Disparities Through Gamification

Through motivating students, encouraging engagement, and advancing fair access to educational resources, gamification has become a promising approach for tackling educational inequities. Gamification can be a significant factor in leveling the playing field and enabling students from a variety of backgrounds to achieve when it comes to entrance exam preparation. Making top-notch educational resources and assistance accessible is one way gamification tackles inequities. Too many kids, especially those from low-income or underprivileged homes, could not have access to typical tutoring or test preparation resources. Regardless of their financial situation, gamified approaches give learners access to interactive and captivating learning experiences at a lower cost and with greater scalability. Gamification helps close the achievement gap between affluent and underprivileged students by democratizing access to educational materials, enabling all students to efficiently prepare for admission tests. Additionally, gamification encourages diversity and tolerance by accommodating a range of learning preferences, methods, and cultural backgrounds. Standardized techniques and rote memorization are frequently prioritized in traditional exam preparation methods, which may not be appealing to all students. Conversely, gamified techniques provide a range of interactive, multimodal learning experiences that are customizable to meet the needs and interests of each individual. Gamification guarantees that learners from a variety of backgrounds feel appreciated, encouraged, and involved in the learning process by offering tailored learning experiences.

Category	**Element Analyzed**	**Frequency (% Relative / % Total)**
Mechanics	Rewards	7 (15.7% / 11.5%)
	Prize	4 (9.3% / 6.3%)
	Achievements	3 (2.7% / 5.7%)
	Points	10 (27.8% / 17.5%)
	Levels	3 (8.3% / 5.3%)
	Badges	7 (19.4% / 12.3%)
	Ranking	6 (26.7% / 20.5%)
Dynamics	Challenge	4 (30.8% / 7.0%)
	Playful activities	2 (15.4% / 4.3%)
	Events	1 (7.8% / 1.7%)
	Tasks	2 (25.4% / 5.3%)
	Roles	1 (7.8% / 1.7%)
	Feedback	1 (6.8% / 1.7%)
	Choices	1 (7.8% / 1.7%)
	Competition	1 (7.8% / 1.7%)
Aesthetics	Narrative	8 (100% / 14.0%)

Gamification can also lessen the negative effects that implicit biases and stereotype threat have on academic achievement. Studies have indicated that stereotype threat, or the anxiety associated with verifying unfavorable preconceptions about one's social group, might impair performance on standardized

assessments, especially for marginalized populations (Steele and Aronson, 1995). In a welcoming and inclusive learning atmosphere, gamified approaches enable learners to take chances, make errors, and grow from their experiences. Gamification allows learners to overcome stereotype threat and realize their full potential by redefining problems as opportunities for development and improvement. Gamification also helps learners feel like they belong and are a part of a community by offering social engagement, peer support, and collaboration options. In particular for students studying independently or remotely, getting ready for an entrance exam can be a lonely and alienating experience. Gamified strategies include social components—like leaderboards, team challenges, and online communities—to foster a sense of unity and support among users. Gamification, especially for students from underrepresented or marginalized groups, helps combat feelings of alienation and isolation by creating a supportive learning community. By facilitating equal access to excellent educational resources, encouraging diversity and inclusivity, lessening the negative effects of stereotype threat, and helping students feel a feeling of community and belonging, gamification has the ability to solve educational inequalities. Gamified approaches present a viable path for educators and policymakers to improve educational equity and social justice, since they empower all learners.

V. Challenges and Considerations

Teachers and legislators face a wide range of obstacles and issues when implementing gamified entrance test preparation strategies, all of which need to be carefully considered in order to assure the programs' efficacy and inclusion. Getting gamified learning exercises precisely in line with the structure and subject matter of the entrance exam is one big problem. Gamification has the potential to increase motivation and engagement, but it also needs to include the material and test-taking techniques that are necessary for passing exams. To guarantee that participants receive thorough preparation, curriculum design and alignment must be carefully considered when balancing gamification aspects with exam-specific knowledge. There are several obstacles to overcome, not only with content alignment but also in creating efficient evaluation systems and giving prompt feedback. To monitor their progress and pinpoint areas for development, participants require honest and helpful comments on their performance. But it takes skill and deliberate thought to create valid and trustworthy assessment instruments that reliably gauge test takers' comprehension of the material while adding gamification features like leaderboards, badges, and points. To maximize the influence of gamified learning activities on learning outcomes, it is imperative to ensure that methods for assessment and feedback are seamlessly integrated into the learning activities. Ensuring inclusion and accessibility for all students, including those with impairments or a range of learning requirements, is another factor to take into account. Accessibility aspects, such supplying support for assistive technology, supporting various learning styles, and offering alternate formats, must be taken into consideration while designing gamified approaches. In addition, it is imperative to tackle possible obstacles to involvement, including restricted internet connectivity or inadequate technological know-how, to guarantee that every student may effectively participate in gamified educational endeavors. Using gamified techniques must also prioritize promoting social justice and equity. Although gamification can help level the playing field by giving everyone equal access to resources and assistance for education, if gamification is not applied carefully, it could exacerbate already-existing inequalities. It is imperative for educators and policymakers to take into account the social and cultural backgrounds of students and guarantee that gamified learning activities are inclusive and applicable to all learners, irrespective of their circumstances or background. To guarantee that gamification helps every student equally, steps should be taken to overcome systemic disparities in access to resources and supportive services. Moreover, the success of educators depends on their ability to support them in using gamified approaches in an effective manner. In order to become acquainted with gamification concepts, technological tools, and pedagogical tactics, educators might need to undergo training and professional development. Teachers should have access to tools and ongoing support to help them incorporate gamified learning activities into their lesson plans and modify them to suit the needs of their students. To maximize their impact on student learning and engagement, gamified approaches must be implemented by instructors who feel competent and confident in their abilities. Applying gamified strategies also involves ethical considerations. It is imperative for educators and legislators to guarantee that gamification tactics are executed in an ethical and responsible manner, upholding the privacy rights, informed consent, and autonomy of participants. Additionally, measures should be taken to reduce the possibility of negative outcomes, such as intense competition, stress, or addiction, and give players' overall growth and wellbeing top priority. Gamified approaches to entrance exam preparation have the ability to completely transform the way we prepare students for success while advancing fairness, inclusion, and social justice in education by carefully and methodically addressing these issues and concerns.

VI. Future Directions and Recommendations

In the future, further study and assessment work should be done to expand on our knowledge of the best practices and efficaciousness of gamified entrance exam preparation methods. RCTs and longitudinal studies can offer important new perspectives on the long-term effects of gamification on information transfer, retention, and learning outcomes. Furthermore, it would be essential to investigate how differently gamified approaches affect different learner populations in order to guarantee fair results in all socioeconomic and cultural circumstances. Increasing personalization and adaptation should be the main emphasis of future efforts to improve the relevance and efficacy of gamified techniques. Learner analytics, individualized feedback systems, and adaptive learning algorithms enable educators to customize learning experiences to each participant's particular requirements, preferences, and stage of development. Additionally, gamified activities will be inclusive and accessible for students with a range of learning styles and abilities if universal design for learning (UDL) principles are incorporated. For gamified approaches to be sustainable and scalable, they must be integrated into current curricula and teaching practices. Teachers ought to investigate how to smoothly include gamified learning activities into lessons, homework, and study sessions on their own. Students will be better prepared for standardized tests and receive reinforcement of academic content when gamified approaches are matched with learning objectives, curriculum standards, and assessment criteria. It is recommended that professional development programs be instituted to provide educators with

the necessary information, skills, and pedagogical methodologies to successfully incorporate gamification into their teaching practices. Communities of practice and ongoing support can help instructors collaborate, share expertise, and troubleshoot together. Gamified methods can be used more successfully and sustainably if educators are equipped with the required training and tools. Stakeholder cooperation is necessary to further gamified entrance exam preparation strategies. Forming alliances with academic institutions, test-prep businesses, and charitable groups can give access to the money, resources, and know-how required to expand gamified programs and reach marginalized communities. Students' opinions and perspectives are recognized during the development process when they are involved in the co-design and evaluation of gamified learning activities. Lastly, it is critical that the design and application of gamified systems give social responsibility and ethical considerations first priority. Maintaining the values of fairness, diversity, openness, and confidentiality is crucial for reducing hazards and unexpected outcomes. Through the integration of ethical and social responsibility into gamified teaching methods, educators may establish secure, encouraging, and empowered learning environments that benefit every student. Future directions and suggestions for gamified entrance exam preparation strategies center on research advancement, improved customization and adaptability, curriculum and instruction integration, educator support, collaboration and partnership development, and social responsibility and ethics as top priorities. Teachers and legislators may use gamification to empower kids, advance educational equity, and get them ready for success on standardized tests and beyond by adopting these suggestions.

VI. Conclusion

To sum up, gamified entrance exam preparation methods present a viable way to solve the problems with standardized testing while fostering student motivation, engagement, and equity in the classroom. Gamification is the practice of transforming exam preparation into an interactive and engaging experience that captures learners' attention and promotes meaningful learning by using game mechanics, social features, and immersive storylines. Gamified techniques have the ability to level the playing field and provide every student the tools they need to succeed by facilitating personalized learning experiences, creating community and cooperation among learners, and giving access to top-notch educational materials. But there are additional issues and concerns with gamified approach execution that need to be properly considered. To optimize the efficacy and impact of gamification in education, it is imperative to guarantee congruence with exam content, furnish efficient systems for assessment and feedback, foster accessibility and inclusion, and offer educators assistance in their implementation. Furthermore, in order to establish inclusive, safe, and supportive learning environments, stakeholders must prioritize social responsibility, ethical considerations, and teamwork. Looking ahead, gamified approaches will need to be further developed and tested in order to reach their full potential as a means of preparation for admission exams. Teachers, legislators, and other stakeholders may use gamification to empower students, advance educational equity, and set them up for success in college and beyond by adopting the recommendations made in this paper. In the end, gamified methods have the power to completely transform the way we think about education and produce more interesting, welcoming, and productive learning environments for all.

References

[1] Smith, J. (2019). The Role of Standardized Tests in College Admissions. Education Policy Analysis Archives, 27(55), 1-24.

[2] Reardon, S. F., & Portilla, X. A. (2016). Recent Trends in Socioeconomic and Racial School Readiness Gaps at Kindergarten Entry. AERA Open, 2(3), 1-18.

[3] Pellegrino, J. W., & Hilton, M. L. (2012). Education for Life and Work: Developing Transferable Knowledge and Skills in the 21st Century. National Academies Press.

[4] Dynarski, S., & Wiederspan, M. (2013). Student Aid Simplification: Looking Back and Looking Ahead. National Tax Journal, 66(4), 889-912.

[5] Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From Game Design Elements to Gamefulness: Defining "Gamification". In Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments (pp. 9-15). ACM.

[6] Gee, J. P. (2007). Good Video Games and Good Learning: Collected Essays on Video Games, Learning, and Literacy. Peter Lang.

[7] Deci, E. L., & Ryan, R. M. (2000). The "What" and "Why" of Goal Pursuits: Human Needs and the Self-Determination of Behavior. Psychological Inquiry, 11(4), 227-268.

[8] Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does Gamification Work? -- A Literature Review of Empirical Studies on Gamification. In Proceedings of the 47th Hawaii International Conference on System Sciences (pp. 3025-3034). IEEE.

[9] Landers, R. N., & Landers, A. K. (2014). An Empirical Test of the Theory of Gamified Learning: The Effect of Leaderboards on Time-on-Task and Academic Performance. Simulation & Gaming, 45(6), 769-785.

[10] Roepke, L. (2015). Gamification of Education: A Review of Literature. In Proceedings of the 5th Annual EuroMed Conference of the EuroMed Academy of Business (pp. 1424-1432). Springer.

[11] Nicholson, S. (2012). A User-Centered Theoretical Framework for Meaningful Gamification. In Proceedings of the Games+Learning+Society 8.0 Conference (pp. 1-16). ETC Press.

[12] Shute, V. J., & Ventura, M. (2013). Stealth Assessment: Measuring and Supporting Learning in Video Games. MIT Press.

[13] Wouters, P., Van Nimwegen, C., Van Oostendorp, H., & Van Der Spek, E. D. (2013). A Meta-Analysis of the Cognitive and Motivational Effects of Serious Games. Journal of Educational Psychology, 105(2), 249-265.

[14] Creswell, J. W., & Plano Clark, V. L. (2017). Designing and Conducting Mixed Methods Research. SAGE Publications.

[15] Denzin, N. K., & Lincoln, Y. S. (2018). The Sage Handbook of Qualitative Research. SAGE Publications.

[16] Levin, H. M., & McEwan, P. J. (2016). Cost-Effectiveness Analysis: Methods and Applications. SAGE Publications.

[17] Vygotsky, L. S. (1978). Mind in Society: The Development of Higher Psychological Processes. Harvard University Press.

[18] Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How Gamification Motivates: An Experimental Study of the Effects of Specific Game Design Elements on Psychological Need Satisfaction. Computers in Human Behavior, 69, 371-380.

[19] Seaborn, K., & Fels, D. I. (2015). Gamification in Theory and Action: A Survey. International Journal of Human-Computer Studies, 74, 14-31.

[20] Landers, R. N. (2014). Developing a Theory of Gamified Learning: Linking Serious Games and Gamification of Learning. Simulation & Gaming, 45(6), 752-768.

[21] Hamari, J., & Koivisto, J. (2015). "Working out for Likes": An Empirical Study on Social Influence in Exercise Games. Computers in Human Behavior, 50, 333-347